

DAWIT TEGEGNWORK WUBALE

SENIOR FULL-STACK DEVELOPER | REACT/NEXT.JS | LARAVEL/NODE.JS | ERP/LIMS

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PROFESSIONAL SUMMARY

Senior-level Full-Stack Engineer with 5 years of experience building, extensively customizing, integrating and supporting production enterprise systems across healthcare, research laboratories, ERP and learning platforms. Served as the primary hands-on developer for a major OpenELIS Global 2 transformation for an Ethiopian research laboratory, working with domain experts, project management and partner teams to convert complex operational workflows into secure software. Strong across React/Next.js, PHP/Laravel, Node.js, Java/Spring, Python, REST APIs, SQL databases, role-based access, reporting, Linux/Docker deployment and production troubleshooting.

TECHNICAL EXPERTISE

Frontend: React, Next.js, JavaScript, TypeScript, HTML5, CSS3, responsive dashboards and complex data-entry workflows

Backend & APIs: PHP, Laravel, Eloquent ORM, migrations, authentication/authorization, queues/jobs, Node.js, Java, Spring Boot, Python, FastAPI, REST APIs and integrations

Enterprise Platforms: Frappe Framework, ERPNext, Odoo, Frappe LMS, OpenELIS Global 2, OpenMRS/Bahmni, InterSystems IRIS for Health

Data & Reporting: PostgreSQL, MySQL/MariaDB, MongoDB, SQL, data modelling, migrations, validation, dashboards and operational reports

Security & Delivery: RBAC, permissions, auditability, Docker, Linux/Ubuntu, Nginx, Git/GitHub, CI/CD, UAT, release support, backups and production stabilization

PROFESSIONAL EXPERIENCE

Full Stack Engineer & Machine Integration Engineer | eHealth IT Service PLC / Orbit Health

Addis Ababa, Ethiopia | Sep 2025–Present

- Build and maintain production healthcare, laboratory and internal operational platforms across backend services, React/JavaScript interfaces, REST APIs, SQL-backed workflows, reports, access controls and deployment support.
- Serve as the primary hands-on developer for extensive OpenELIS Global 2 customization for AHRI, adapting a clinical LIMS into a research-laboratory platform while collaborating with laboratory domain experts, the project manager and technical partner teams.
- Implemented research and pathology-style workflows covering sample creation, quality control, slide preparation, staining, microscopy/diagnosis, storage and inventory, reporting and performance tracking, including workflow states, validation rules and role-specific actions.
- Build and customize Frappe/ERPNext, Odoo and Frappe LMS solutions, including DocTypes/data models, roles and permissions, forms, approvals, dashboards, reports, learning workflows and integration points.
- Develop and troubleshoot Laravel/PHP, Java, Python and JavaScript logic, SQL queries, APIs, machine/data integrations, Docker/Linux environments, release testing and live production issues.
- Translate requirements from hospital and laboratory users into technical designs and tested features; coordinate decisions, UAT, documentation, rollout follow-up and defect resolution with developers, managers and domain stakeholders.

Software Engineer | iMedway

Remote / Addis Ababa, Ethiopia | Sep 2024–Aug 2025

- Developed and localized hospital-management modules and database-backed workflows for patient administration, clinical records, reporting, billing/payment and operational support using web technologies, APIs and InterSystems IRIS for Health.
- Built and supported forms, dashboards, REST API-backed features and integration workflows; investigated defects across UI behaviour, API responses, mappings, database records, permissions and reports.
- Worked directly with product, clinical and technical stakeholders to clarify requirements, validate workflows, document changes and support production releases.

Software Engineer | Defense Specialized Referral Hospital

Bishoftu, Ethiopia | Sep 2021–Aug 2024

- Developed, customized and supported mission-critical hospital information workflows across registration, EMR/LIS processes, user access, SQL-backed reporting and day-to-day operational data.
- Resolved application, database, workflow and data-quality issues in a live multi-user environment where uptime, confidentiality, accurate permissions and responsive support were essential.
- Partnered with clinicians, laboratory professionals, administrators and technical teams during requirements analysis, implementation, testing, deployment, training, documentation and production support.

SELECTED ENTERPRISE DELIVERY

AHRI Research Laboratory Information System | *Most complex project* | *OpenELIS Global 2*

- Challenge: a general clinical laboratory platform could not represent the institute's research and pathology processes, multi-step sample lifecycle, quality controls, specialist decisions, storage needs and management reporting.
- Ownership: served as the primary hands-on developer and owned a substantial share of full-stack implementation and redesign across workflow behaviour, data-backed forms, validation, permissions, dashboards and reports.
- Delivery model: domain experts and project leadership defined and validated laboratory practice; I translated those requirements into architecture, application behaviour, workflow states and deployable software, then iterated through review, UAT and production follow-up.

Internal ERP & Learning Platforms | *Frappe / ERPNext / Odoo / Frappe LMS*

- Extended enterprise workflows for users, roles, applications, approvals, inventory and operational processes, learning content, assignments, progress, dashboards and reporting.
- Worked across DocTypes/data models, Python and JavaScript logic, roles/permissions, client forms, reports, APIs, MariaDB/MySQL data, deployment checks and production troubleshooting.

Laravel Enterprise Applications | *PHP / Laravel / REST APIs / SQL*

- Built Laravel-based business and operational applications using REST APIs, service-layer patterns, Eloquent ORM, migrations, validation, queues/jobs, authentication, role-based authorization and MySQL/PostgreSQL.
- Designed maintainable modules for complex forms, approvals, dashboards, reporting and external-service integrations; supported Linux/Docker deployment and production issue resolution.

Hospital Payment & External-System Integrations | *APIs, middleware and transaction workflows*

- Contributed to payment and cross-system integration workflows involving API communication, structured payloads, transaction validation, status handling, failure investigation and reconciliation-oriented reporting.
- Applied defensive integration design around asynchronous responses, duplicate or failed transactions, status visibility, error tracing and supportability rather than assuming every external request succeeds.

Healthcare, Laboratory & Machine Integrations | *EMR/LIS, analyzers and imaging workflows*

- Worked with OpenMRS/Bahmni and OpenELIS order/result flows, laboratory analyzers and structured health-data exchange; investigated mismatches across devices, APIs, mappings, database records and user workflows.
- Supported radiology/PACS workflow integration using Orthanc concepts, including patient/order matching, study availability, configuration and connectivity troubleshooting.

ARCHITECTURE & DELIVERY STRENGTHS

- Enterprise architecture: modular Laravel and service-based workflows, REST API contracts, relational data modelling, multi-user dashboards, reporting and integration boundaries.
- Security and control: role-based access, permission scoping, validation, auditability, controlled handling of sensitive operational data and production issue diagnosis.
- Operational reliability: Linux/Docker deployment support, Nginx and configuration checks, backups, logs, database inspection, release validation and post-deployment stabilization.
- Product delivery: requirements discovery, process mapping, stakeholder alignment, UAT, defect prioritization, technical documentation, user training and iterative rollout support.

EDUCATION

BSc in Computer Science and Engineering

Adama Science and Technology University, Ethiopia | Sep 2017–Sep 2021

LANGUAGES

Amharic: Native | **English:** Professional working proficiency